Clustering

Basics

- unsupervised learning - we have data without labels
- **cluster** - a set of similar objects grouped together
- goal - high similarity within a cluster, but low between clusters

Clustering quality

- **silhouette coefficient** - for i element:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

  - $a(i)$ - mean distance to other samples in the same cluster
  - $b(i)$ - minimum mean distance to samples in other clusters
  - value close to 1 - element well assigned
  - value close to 0 - lies on the boundry between clusters
  - negative - missasigned

Clustering method

- **k-means algorithm**:
  1. choose k initial centroids
  2. assign each point to the nearest centroid
  3. recompute each centroid as the mean of it's assigned points
  4. if any centroid changed, go back to step 2
- problems:
  - bad initialization
  - empty clusters
  - result depends on the metric